Mat-Bio Exchange: Turing Patterns

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For these exercises, we will be using two programs: Firstly, to analyse the reaction part of the models we will be using RStudio again (either running on your laptop, downloaded for free from https://posit.co/downloads/, or running online through https://posit.cloud, see the notes from yesterday). Secondly, to run the full spatial simulations, we will be using VisualPDE, a very powerful yet simple to use online PDE solver service.

1 Turing Patterns: stripes and spots

We will be using the model for the formation of Turing patterns that was used by Murray (2003) to describe animal coat patterns, as was shown on the slides. Murray has chosen this model because it is a real experimental system with parameters associated with real kinetics based on a specific reaction involving the substrates oxygen and uric acid which react in the presence of the enzyme uricase. The dimensionless form of the empirical rate equations for the oxygen (v) and the uric acid (u) is shown below:

$$\frac{\partial u}{\partial t} = \gamma f(u,v) + \nabla^2 u\,, \tag{1}$$

$$\frac{\partial v}{\partial t} = \gamma g(u,v) + D_v \nabla^2 v\,, \tag{2}$$

in which

$$f(u,v) = a - u - h(u,v)\,, \qquad g(u,v) = \alpha(b-v) - h(u,v)\,,$$

$$h(u,v) = \frac{\rho u v}{1 + u + K u^2}\,.$$

Basically, u and v are supplied at constant rates a and αb, degrade linearly proportional to their concentrations and both are used up in a reaction at a rate $h(u,v)$. The form of $h(u,v)$ exhibits so-called substrate inhibition: For small u, $h(u,v)$ increases with u, but for large u it decreases with u. This is what is meant by substrate inhibition. The parameter K is a measure of the severity of the inhibition. Thus, the uptake rate is like a Michaelis–Menten form for small u, reaches a maximum at $u = 1/\sqrt{K}$, and then decreases with increasing u.

1. Use RStudio to analyse the ODE version of the model, in which we ignore the diffusion part of Eq. 1 and Eq. 2. Open `turing.R` in RStudio, which uses `GrindR` to analyse the equations. Run step by step each command. First the model is defined,

then the parameter values and initial state. run() creates a timeplot, and plane() draws a phase plane. Please pay attention to the arguments of those functions, which are explained in the comments.

2. Determine by hand the graphical Jacobian. Is it possible to know the stability of the equilibrium using the graphical Jacobian? Numerically determine the stability and Jacobian, using newton().

3. Identify which variable is the activator and which variable is the inhibitor. Which variable has to diffuse faster in order to make a Turing pattern possible?

4. Identify in which of the two classes of Turing patterns this model falls. Will a peak in the activator correspond to a peak in the inhibitor or with a trough in the inhibitor? Can you intuitively understand why?

5. Numerically explore the spatial patterns that can be formed by going to the following website: https://visualpde.com/sim/?mini=bbzCal5o Initially u is displayed, with the concentrations depicted following the colourbar shown at the bottom, scaled between the minimum and maximum value within the field. To see instead the distribution of v, click on the bottom-left stack symbol and click on v. Note that the colourbar below will then also automatically change to v. Press the "play" symbol on the left to let the simulation evolve. Clicking into the field will locally increase the concentration of u. Use this to explore how the pattern dynamically changes. You can reset the simulation by pressing the "return" button on the left. What pattern do you observe? Are peaks in the activator corresponding to peaks in the inhibitor or troughs in the inhibitor? Does that correspond to your expectation?

6. Parameters can be changed by clicking on the top-left "$f(x)$" button, and then clicking on "Parameters". First, vary γ. How does this change the pattern? Can you understand these changes by looking at the equations? You can see the equations at the top of the pop-out panel.

7. Next, vary all other parameters. (This can be done for different values of γ.) How does the pattern change? Can you get 3 qualitatively different patterns?

8. Using `turing.R`, observe how the nullclines change under such changes. Formulate an intuitive idea how changes in the pattern can be understood from the changes in the nullclines.

9. Vary the diffusion constants and get a feeling how they affect the patterning. To change the diffusion constants, choose "Edit" under the "$f(x)$" button.

10. From today's lecture notes, calculate the expected most unstable wave length and compare this to the simulation results. Code to do so is provided in turing_wavelength.R. Note that the most unstable wavelength close to the equilibrium cannot fully predict the eventually established wavelength far from equilibrium, which depends on nonlinear effects as well as field size constraints.

11. Make your own animal shapes, and explore the effects that Turing patterns can generate! To do so, click on the "tool" symbol on the right, click on "Images", and then click on the image of Turing. It now allows to replace the image of Turing by another image that you can upload. The provided files `hydra.png`, `fur.png` and `puma.png`, are examples that I made, but it is of course much more

interesting to generate your own examples or find funny pictures on the internet. At the moment everything darker will be used as the domain for the simulations, but let me know if you want something more fancy. Also, the vectorial file `hydra.svg` is provided to straightforwardly change the layout, for example by using the open source software Inkscape. Now click on "Domain" and then on "Custom", and the domain will be constrained to the provided picture. Moreover, if you want to change the colours of the simulation, click on the bottom-left stack symbol, click on "Edit view", and then click on "Colour".

12. The self-organising patterning process in Hydra could be linked to Turing patterns along two axes of organisation: the longitudinal pattern regulating the main body axis, and the periodic radial patterning in the head region regulating the tentacle formation. Choose "Boundary Conditions" under the "$f(x)$" button, and select mixed boundary conditions for both variables. Make the left and right boundary periodic and the top and bottom boundary condition zero-flux (Neumann), such that the simulation represents a folded-out Hydra. Can you get a right patterning, in which the radial patterning is restricted to the apical region? You might want to consider i) replacing the equations by the Gierer-Meinhardt equations presented by Szymon; ii) adding additional spatial dependency in the equations (presumably due to the earlier initial longitudinal patterning), which can be done by introducing x and/or y at any place in the equations (making the rate of processes explicitly dependent on the spatial position), for example by replacing a by $a * (1 + 0.01 * x)$; and/or iii) extending the model by doubling the equations to four equations, and then introducing different parameters for each set and/or some level of cross-talk (or using the Gierer-Meinhardt model for the second set of equations). Note that the number of species can be changed under the "$f(x)$" button, "Advanced options". Set "# Species" to 4, introduce names for the new Species, and then double the equations, paying attention to use the correct variable names.

Suggested Reading

B. Ermentrout. Stripes and spots? Nonlinear effects in bifurcation of reaction-diffusion equations on the square. *Proc. R. Soc. Lond. A*, 434:413–417 (1991). Mathematical explanation of the difference between stripes and spots.

H. Meinhardt. Complex pattern formation by a self-destabilization of established patterns: chemotactic orientation and phyllotaxis as examples. *C. R. Biol.*, 326(2):223–237 (Feb 2003). Overview of the wide range of patterning found by Hans Meinhardt.

J. D. Murray. *Mathematical Biology II: Spatial models and biomedical applications.* Springer-Verlag, Berlin, 3rd edn. (2003). Classical book describing, among others, the animal fur patterns.

A. M. Turing. The chemical basis of morphogenesis. *Phil. Trans. Roy. Soc. Lond. B*, 237:37–72 (1952). Turing's original paper.